

**LAPORAN WORKSHOP
KECERDASAN BUATAN**



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2026

Analisa Output Bab 4

➤ Example :

Example 4.1

```
Problems Output Terminal Ports Debug Console
PS C:\Users\ayu zenska\OneDrive\AI BAB 4> & "C:\Users\ayu zenska\AppData\Local\Microsoft\WindowsApps\python3.9.exe" "C:\Users\ayu zenska\OneDrive\AI BAB 4/example 4_1.py"
[6.9958177]
PS C:\Users\ayu zenska\OneDrive\AI BAB 4>
```

Program menggunakan single neuron untuk mempelajari logika OR. Hasil menunjukkan bahwa neuron dapat mempelajari hubungan antara input dan output melalui proses iterasi.

Example 4.2

```
PS C:\Users\ayu zenska\OneDrive\AI BAB 4> & "C:\Users\ayu zenska\AppData\Local\Microsoft\WindowsApps\python3.9.exe" "C:\Users\ayu zenska\OneDrive\AI BAB 4/example 4_2.py"
[1 0]
[(2, 5), (5, 2), (2, 1)]
PS C:\Users\ayu zenska\OneDrive\AI BAB 4>
```

Program mengembangkan model dengan tiga input menggunakan beberapa neuron. Hasilnya menunjukkan bahwa jaringan dapat memproses lebih banyak fitur untuk menghasilkan output.

Example 4.3

```
PS C:\Users\ayu zenska\OneDrive\AI BAB 4> & "C:\Users\ayu zenska\AppData\Local\Microsoft\WindowsApps\python3.9.exe" "C:\Users\ayu zenska\OneDrive\AI BAB 4/example 4_3.py"
2026-09-30 19:45:36.163974: I tensorflow/core/util/port.cc:155] oneDNN custom operations are on. You may see slightly different numerical results due to floating-point round-off errors from different computation orders. To turn them off, set the environment variable 'TF_ENABLE_ONEDNN_OPTS=0'.
2026-09-30 19:45:42.561822: I tensorflow/core/util/port.cc:153] oneDNN custom operations are on. You may see slightly different numerical results due to floating-point round-off errors from different computation orders. To turn them off, set the environment variable 'TF_ENABLE_ONEDNN_OPTS=0'.
C:\Users\ayu zenska\AppData\Local\Packages\PythonSoftwareFoundation.Python.3.9_qbz5n2kfra8p0\LocalCache\local-packages\Python39\site-packages\Keras\src\layers\core\dense.py:93: UserWarning: Do not pass an 'input_shape' / 'input_dim' argument to a layer. When using Sequential models, prefer using an 'Input(shape)' object as the first layer in the model instead.
  super().__init__(activity_regularizer=activity_regularizer, **kwargs)
2026-09-30 19:45:48.877435: I tensorflow/core/platform/cpu_feature_guard.cc:210] This TensorFlow binary is optimized to use available CPU instructions in performance-critical operations.
To enable the following instructions: SSE3 SSE4.1 SSE4.2 AVX AVX2 FMA, in other operations, rebuild TensorFlow with the appropriate compiler flags.
Model: "sequential"

Layer (type)                Output Shape                Param #
-----
dense (Dense)                (None, 10)                  310
dense_1 (Dense)              (None, 10)                  110
dense_2 (Dense)              (None, 1)                   11

Total params: 431 (1.68 KB)
Trainable params: 431 (1.68 KB)
```

Model MLP digunakan untuk melakukan klasifikasi pada dataset breast cancer. Jaringan mampu mempelajari pola dari beberapa fitur input untuk menghasilkan prediksi.

Example 4.4

the first layer in the model instead.

```
super().__init__(Activity_regularizer=activity_regularizer, **kwargs)
2020-08-10 15:44:39.424782: I tensorflow/core/platform/cpu_feature_guard.cc:210] This TensorFlow binary is optimized to use available CPU instructions in performance-critical operations.
To enable the following instructions: SSE3 SSE4.1 SSE4.2 AVX AVX2 FMA, in other operations, rebuild TensorFlow with the appropriate compiler flags.
Model: "sequential"
```

Layer (type)	Output Shape	Param #
conv2d (Conv2D)	(None, 30, 30, 6)	66
average_pooling2d (AveragePooling2D)	(None, 15, 15, 6)	0
conv2d_1 (Conv2D)	(None, 13, 13, 16)	886
average_pooling2d_1 (AveragePooling2D)	(None, 6, 6, 16)	0
flatten (Flatten)	(None, 576)	0
dense (Dense)	(None, 128)	69,768
dense_1 (Dense)	(None, 84)	19,164
dense_2 (Dense)	(None, 10)	850

Total params: 81,194 (317.16 KB)
Trainable params: 81,194 (317.16 KB)
Non-trainable params: 0 (0.00 B)
PS C:\Users\ayu_zenska\OneDrive\AI&S&B 4>

Model MLP dimodifikasi dengan beberapa layer Dense. Penambahan layer membuat jaringan dapat mempelajari hubungan yang lebih kompleks dari data.

Example 4.5

File Edit Selection View Go Help

example 4.3.py example 4.4.py example 4.5.py example 4.6.py example 4.7.py

example 4.5.py

```
1 # Example 4.5 AlexNet with keras
2
3 import keras
4 from keras.models import Sequential
```

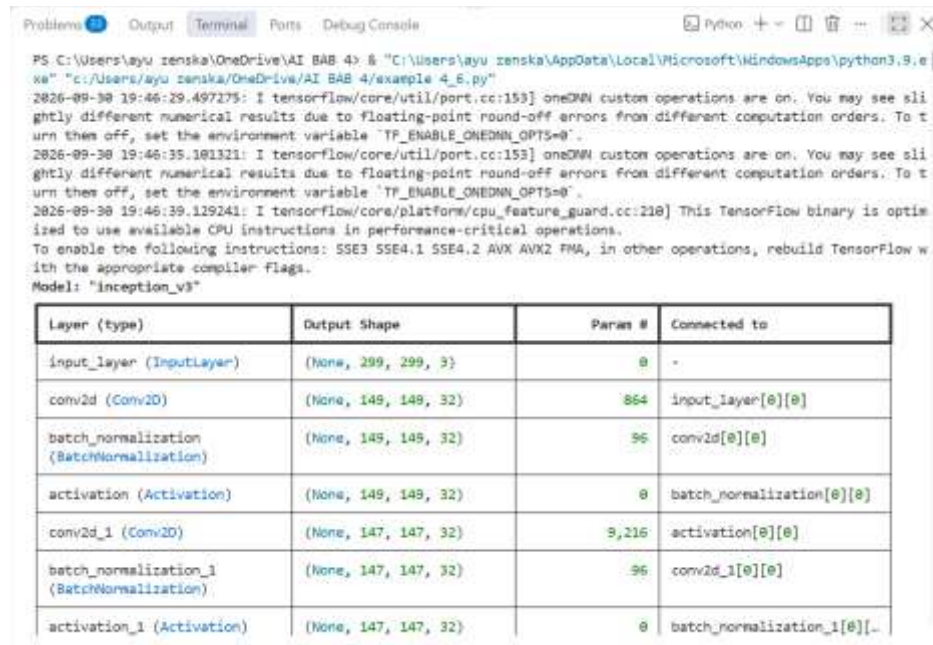
Problems Output Results Ports Debug Console

python

Layer (type)	Output Shape	Param #
conv2d (Conv2D)	(None, 54, 54, 96)	34,344
max_pooling2d (MaxPooling2D)	(None, 27, 27, 96)	0
conv2d_1 (Conv2D)	(None, 17, 17, 256)	2,879,808
max_pooling2d_1 (MaxPooling2D)	(None, 8, 8, 256)	0
conv2d_2 (Conv2D)	(None, 8, 8, 384)	881,120
conv2d_3 (Conv2D)	(None, 4, 4, 384)	1,327,488
conv2d_4 (Conv2D)	(None, 2, 2, 256)	884,000
max_pooling2d_2 (MaxPooling2D)	(None, 1, 1, 256)	0
flatten (Flatten)	(None, 256)	0
dense (Dense)	(None, 4096)	1,052,072
dropout (Dropout)	(None, 4096)	0

Program memperkenalkan arsitektur AlexNet untuk klasifikasi gambar. Model menggunakan beberapa convolutional layer dan fully connected layer untuk mengenali objek.

Example 4.6



PS C:\Users\ayu zenska\OneDrive\AI BAB 4> & "C:\Users\ayu zenska\AppData\Local\Microsoft\WindowsApps\python3.9.exe" "c:\Users\ayu zenska\OneDrive\AI BAB 4\example 4_6.py"

2026-09-30 19:46:29.497275: I tensorflow/core/util/port.cc:153] oneDNN custom operations are on. You may see slightly different numerical results due to floating-point round-off errors from different computation orders. To turn them off, set the environment variable 'TF_ENABLE_ONEDNN_OPTS=0'.

2026-09-30 19:46:35.101321: I tensorflow/core/util/port.cc:153] oneDNN custom operations are on. You may see slightly different numerical results due to floating-point round-off errors from different computation orders. To turn them off, set the environment variable 'TF_ENABLE_ONEDNN_OPTS=0'.

2026-09-30 19:46:39.129241: I tensorflow/core/platform/cpu_feature_guard.cc:210] This TensorFlow binary is optimized to use available CPU instructions in performance-critical operations.

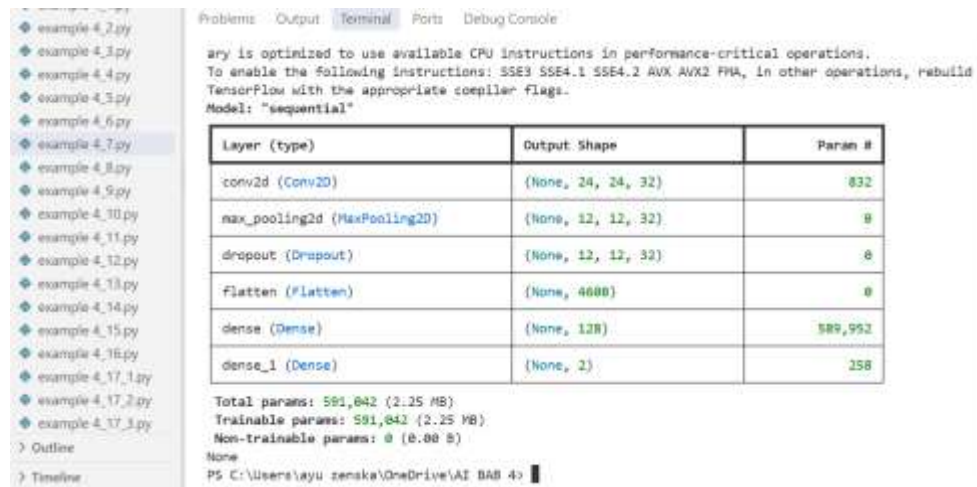
To enable the following instructions: SSE3 SSE4.1 SSE4.2 AVX AVX2 FMA, in other operations, rebuild TensorFlow with the appropriate compiler flags.

Model: "inception_v3"

Layer (type)	Output Shape	Param #	Connected to
input_layer (InputLayer)	(None, 299, 299, 3)	0	-
conv2d (Conv2D)	(None, 149, 149, 32)	864	input_layer[0][0]
batch_normalization (BatchNormalization)	(None, 149, 149, 32)	96	conv2d[0][0]
activation (Activation)	(None, 149, 149, 32)	0	batch_normalization[0][0]
conv2d_1 (Conv2D)	(None, 147, 147, 32)	9,216	activation[0][0]
batch_normalization_1 (BatchNormalization)	(None, 147, 147, 32)	96	conv2d_1[0][0]
activation_1 (Activation)	(None, 147, 147, 32)	0	batch_normalization_1[0][0]

InceptionV3 digunakan sebagai model CNN untuk klasifikasi gambar. Arsitektur ini memiliki banyak layer sehingga mampu mengekstraksi berbagai fitur dari gambar.

Example 4.7



any is optimized to use available CPU instructions in performance-critical operations.

To enable the following instructions: SSE3 SSE4.1 SSE4.2 AVX AVX2 FMA, in other operations, rebuild TensorFlow with the appropriate compiler flags.

Model: "sequential"

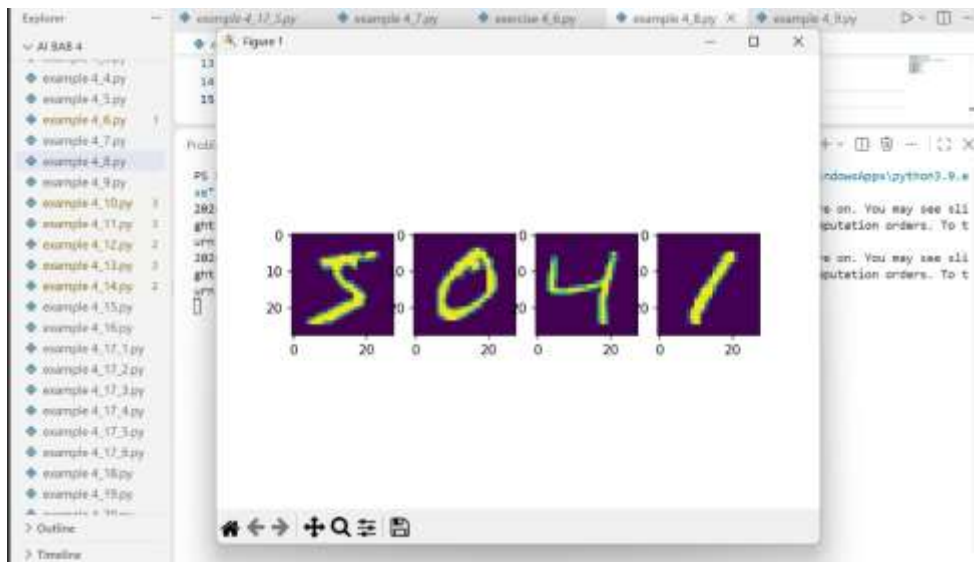
Layer (type)	Output Shape	Param #
conv2d (Conv2D)	(None, 24, 24, 32)	832
max_pooling2d (MaxPooling2D)	(None, 12, 12, 32)	0
dropout (Dropout)	(None, 12, 12, 32)	0
Flatten (Flatten)	(None, 4608)	0
dense (Dense)	(None, 128)	589,952
dense_1 (Dense)	(None, 2)	258

Total params: 591,042 (2.25 MB)
Trainable params: 591,042 (2.25 MB)
Non-trainable params: 0 (0.00 B)
None

PS C:\Users\ayu zenska\OneDrive\AI BAB 4>

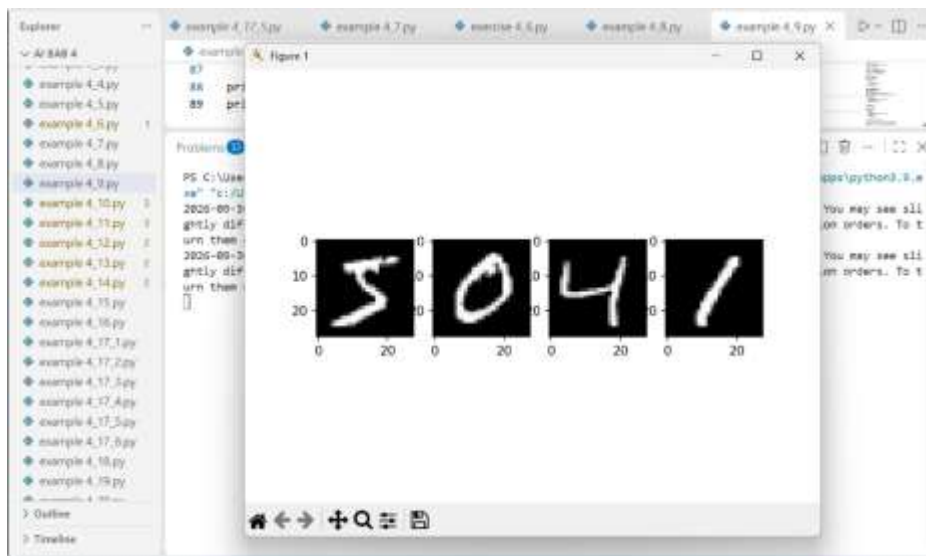
Program membuat CNN sederhana dengan convolution, pooling, dan dense layer. CNN digunakan untuk mempelajari fitur gambar secara bertahap.

Example 4.8



CNN dikembangkan dengan menambahkan layer untuk meningkatkan proses ekstraksi fitur. Hasil menunjukkan bahwa struktur jaringan dapat memengaruhi kemampuan model dalam mengenali pola.

Example 4.9



CNN digunakan untuk klasifikasi dataset MNIST. Model belajar mengenali pola angka berdasarkan bentuk dan karakteristik piksel gambar.

Example 4.10



Model: "vgg16"

Layer (type)	Output Shape	Param #
input_layer (InputLayer)	(None, 224, 224, 3)	0
block1_conv1 (Conv2D)	(None, 224, 224, 64)	1,792
block1_conv2 (Conv2D)	(None, 224, 224, 64)	36,928
block1_pool1 (MaxPooling2D)	(None, 112, 112, 64)	0
block2_conv1 (Conv2D)	(None, 112, 112, 128)	73,856
block2_conv2 (Conv2D)	(None, 112, 112, 128)	147,584
block2_pool1 (MaxPooling2D)	(None, 56, 56, 128)	0
block3_conv1 (Conv2D)	(None, 56, 56, 256)	295,168
block3_conv2 (Conv2D)	(None, 56, 56, 256)	590,080
block3_conv3 (Conv2D)	(None, 56, 56, 256)	390,080

Program menggunakan VGG19 dan menampilkan struktur model. Hal ini memperlihatkan banyaknya layer yang digunakan VGG19 dalam proses ekstraksi fitur.

Example 4.11

```
PS C:\Users\ayu zenska\OneDrive\AI BAB 4> & "C:\Users\ayu zenska\AppData\Local\Microsoft\WindowsApps\python3.9.exe" "c:/Users/ayu zenska/OneDrive/AI BAB 4/example_4_11.py"
2026-09-30 19:57:49.404444: I tensorflow/core/util/port.cc:153] oneDNN custom operations are on. You may see slightly different numerical results due to floating-point round-off errors from different computation orders. To turn them off, set the environment variable 'TF_ENABLE_ONEDNN_OPTS=0'.
2026-09-30 19:57:53.869046: I tensorflow/core/util/port.cc:153] oneDNN custom operations are on. You may see slightly different numerical results due to floating-point round-off errors from different computation orders. To turn them off, set the environment variable 'TF_ENABLE_ONEDNN_OPTS=0'.
2026-09-30 19:57:56.681224: I tensorflow/core/platform/cpu_feature_guard.cc:210] This TensorFlow binary is optimized to use available CPU instructions in performance-critical operations.
To enable the following instructions: SSE3 SSE4.1 SSE4.2 AVX AVX2 FMA, in other operations, rebuild TensorFlow with the appropriate compiler flags.
1/1 ----- 1s 960ms/step
Downloading data from https://storage.googleapis.com/download.tensorflow.org/data/imagenet_class_index.json
35363/35363 ----- 1s 31us/step
[[('n01871265', 'tusk', 0.46696895), ('n02504458', 'African_elephant', 0.38354492), ('n02504013', 'Indian_elephant', 0.12821345), ('n02088466', 'bloodhound', 0.0048397733), ('n02109047', 'Great_Dane', 0.004292182)]]
PS C:\Users\ayu zenska\OneDrive\AI BAB 4> █
```

VGG16 digunakan untuk melakukan klasifikasi gambar Elephant.jpg. Model menghasilkan beberapa kemungkinan kelas beserta nilai confidence untuk setiap kelas.

Example 4.12



```
3 from keras.applications.imagenet_utils import decode_predictions
4 import numpy as np
```

ilenet_1_0_224_tf.h5
17225924/17225924 43s 2us/step
Model: "mobilenet_1_00_224"

Layer (type)	Output Shape	Param #
input_layer (InputLayer)	(None, 224, 224, 3)	0
conv1 (Conv2D)	(None, 112, 112, 32)	864
conv1_bn (BatchNormalization)	(None, 112, 112, 32)	128
conv1_relu (ReLU)	(None, 112, 112, 32)	0
conv_dw_1 (DepthwiseConv2D)	(None, 112, 112, 32)	288
conv_dw_1_bn (BatchNormalization)	(None, 112, 112, 32)	128
conv_dw_1_relu (ReLU)	(None, 112, 112, 32)	0
conv_pw_1 (Conv2D)	(None, 112, 112, 64)	2,048
conv_pw_1_bn (BatchNormalization)	(None, 112, 112, 64)	256

Program melakukan klasifikasi menggunakan model CNN yang berbeda. Setiap model menghasilkan prediksi berdasarkan fitur yang dipelajari dari gambar.

Example 4.13



```
3 from keras.applications.imagenet_utils import decode_predictions
4 import numpy as np
```

Model: "vgg16"

Layer (type)	Output Shape	Param #
input_layer (InputLayer)	(None, 224, 224, 3)	0
block1_conv1 (Conv2D)	(None, 224, 224, 64)	1,792
block1_conv2 (Conv2D)	(None, 224, 224, 64)	36,928
block1_pool (MaxPooling2D)	(None, 112, 112, 64)	0
block2_conv1 (Conv2D)	(None, 112, 112, 128)	73,656
block2_conv2 (Conv2D)	(None, 112, 112, 128)	147,584
block2_pool (MaxPooling2D)	(None, 56, 56, 128)	0
block3_conv1 (Conv2D)	(None, 56, 56, 256)	295,168
block3_conv2 (Conv2D)	(None, 56, 56, 256)	590,880
block3_conv3 (Conv2D)	(None, 56, 56, 256)	590,880
block3_pool (MaxPooling2D)	(None, 28, 28, 256)	0

Model VGG16 digunakan bersama webcam untuk melakukan klasifikasi secara langsung. Program menunjukkan penerapan CNN pada input gambar secara real-time.

Example 4.14



Layer (type)	Output Shape	Param #	Connected to
input_layer (InputLayer)	(None, 224, 224, 3)	0	-
conv1_pad (ZeroPadding2D)	(None, 230, 230, 3)	0	input_layer[0][0]
conv1_conv (Conv2D)	(None, 112, 112, 64)	9,472	conv1_pad[0][0]
conv1_bn (BatchNormalization)	(None, 112, 112, 64)	256	conv1_conv[0][0]
conv1_relu (Activation)	(None, 112, 112, 64)	0	conv1_bn[0][0]
pool1_pad (ZeroPadding2D)	(None, 114, 114, 64)	0	conv1_relu[0][0]
pool1_pool (MaxPooling2D)	(None, 56, 56, 64)	0	pool1_pad[0][0]
conv2_block1_conv (Conv2D)	(None, 56, 56, 64)	4,160	pool1_pool[0][0]

Program menggunakan beberapa arsitektur CNN untuk klasifikasi melalui webcam. Hasil prediksi dapat berbeda karena setiap model memiliki arsitektur dan karakteristik pembelajaran yang berbeda.

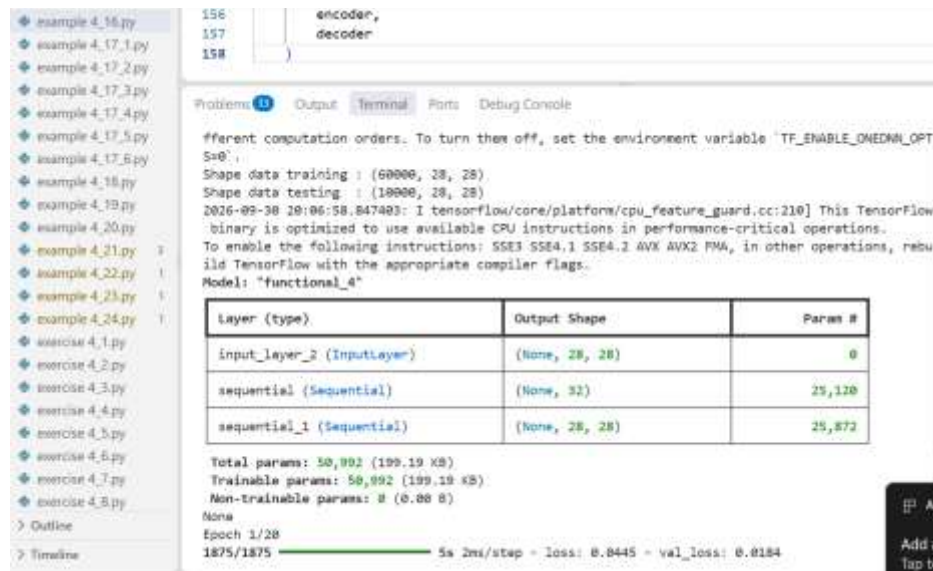
Example 4.15



Layer (type)	Output Shape	Param #
input_layer (InputLayer)	(None, 224, 224, 3)	0
block1_conv1 (Conv2D)	(None, 224, 224, 64)	1,792
block1_conv2 (Conv2D)	(None, 224, 224, 64)	36,928
block1_pool (MaxPooling2D)	(None, 112, 112, 64)	0
block2_conv1 (Conv2D)	(None, 112, 112, 128)	73,856
block2_conv2 (Conv2D)	(None, 112, 112, 128)	147,584
block2_pool (MaxPooling2D)	(None, 56, 56, 128)	0
block3_conv1 (Conv2D)	(None, 56, 56, 256)	295,168
block3_conv2 (Conv2D)	(None, 56, 56, 256)	590,080
block3_conv3 (Conv2D)	(None, 56, 56, 256)	590,080

Autoencoder digunakan untuk melakukan kompresi dan rekonstruksi gambar MNIST. Gambar dikurangi menjadi representasi yang lebih kecil kemudian direkonstruksi kembali.

Example 4.16



```
156 encoder,
157 decoder
158 )
```

Problems 13 Output Terminal Ports Debug Console

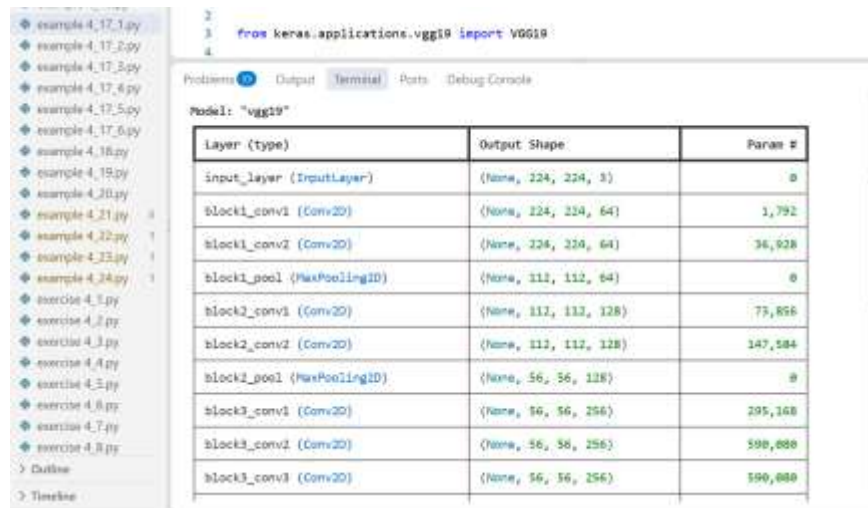
fferent computation orders. To turn them off, set the environment variable 'TF_ENABLE_ONEDNN_OPTS=0'.
Shape data training : (60000, 28, 28)
Shape data testing : (10000, 28, 28)
2026-09-30 10:05:58.847403: I tensorflow/core/platform/cpu_feature_guard.cc:210] This TensorFlow binary is optimized to use available CPU instructions in performance-critical operations.
To enable the following instructions: SSE3 SSE4.1 SSE4.2 AVX AVX2 FMA, in other operations, rebuild TensorFlow with the appropriate compiler flags.
Model: "functional_4"

Layer (type)	Output Shape	Param #
input_layer_2 (InputLayer)	(None, 28, 28)	0
sequential (Sequential)	(None, 32)	25,120
sequential_1 (Sequential)	(None, 28, 28)	25,872

Total params: 50,992 (199.19 KB)
Trainable params: 50,992 (199.19 KB)
Non-trainable params: 0 (0.00 KB)
None
Epoch 1/20
1875/1875 — 5s 2ms/step - loss: 0.8445 - val_loss: 0.0184

Program memperlihatkan proses pembelajaran model melalui visualisasi hasil rekonstruksi. Semakin kecil nilai loss, semakin dekat hasil rekonstruksi dengan gambar asli.

Example 4.17

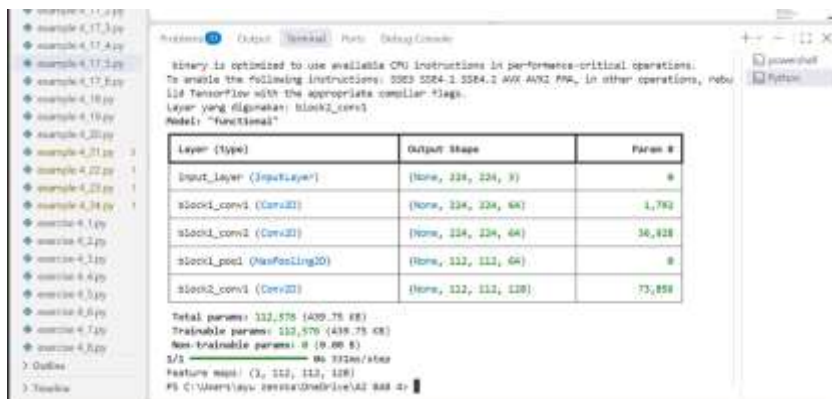


```
2
3 from keras.applications.vgg19 import VGG19
4
```

Problems 13 Output Terminal Ports Debug Console

Model: "vgg19"

Layer (type)	Output Shape	Param #
input_layer (InputLayer)	(None, 224, 224, 3)	0
block1_conv1 (Conv2D)	(None, 224, 224, 64)	1,792
block1_conv2 (Conv2D)	(None, 224, 224, 64)	36,928
block1_pool (MaxPooling2D)	(None, 112, 112, 64)	0
block2_conv1 (Conv2D)	(None, 112, 112, 128)	73,856
block2_conv2 (Conv2D)	(None, 112, 112, 128)	147,584
block2_pool (MaxPooling2D)	(None, 56, 56, 128)	0
block3_conv1 (Conv2D)	(None, 56, 56, 256)	295,168
block3_conv2 (Conv2D)	(None, 56, 56, 256)	590,880
block3_conv3 (Conv2D)	(None, 56, 56, 256)	590,880



Problems Output Terminal Ports Debug Console

```

2024-09-20 20:12:39.758888: I tensorflow/core/platform/cpu_feature_guard.cc:210] This TensorFlow binary is optimized to use available CPU instructions in performance-critical operations.
To enable the following instructions: SSE3 SSE4.1 SSE4.2 AVX AVX2 FMA, in other operations, rebuild TensorFlow with the appropriate compiler flags.
Model: "functional"

```

Layer (type)	Output Shape	Param #
Input_layer (InputLayer)	(None, 224, 224, 3)	0
block1_conv1 (Conv2D)	(None, 224, 224, 64)	1,792

```

Total params: 1,792 (7.00 KB)
Trainable params: 1,792 (7.00 KB)
Non-trainable params: 0 (0.00 B)
1/1 [-----] 0s 181ms/step
Feature maps: (3, 224, 224, 64)
Traceback (most recent call last):
  File "C:\Users\ayu\OneDrive\AI\BAG 4\example_4_17_8.py", line 38, in module
    plot_feature_maps(feature_maps)
  NameError: name 'plot_feature_maps' is not defined
PS C:\Users\ayu\OneDrive\AI\BAG 4>

```

VGG19 digunakan untuk memvisualisasikan filter dan feature maps. Visualisasi menunjukkan bahwa CNN dapat mendeteksi pola seperti garis, tepi, dan tekstur pada gambar.

Example 4.18

Problems Output Terminal Ports Debug Console

```

[[0.09078972]
 [0.6352919 ]
 [0.0060996 ]
 [0.04160859 ]
 [0.77408285]]

[[0.00277904]
 [0.4583376 ]
 [0.2783484 ]
 [0.5389115 ]
 [0.6834888 ]]

[[0.8743023 ]
 [0.40529555 ]
 [0.4130154 ]
 [0.02873584 ]
 [0.21470553]]]

tf.Tensor(
[[ 0.7205539 -0.10330673 -0.2503899 -0.24531866]
 [ 0.59569645 0.4957159 -0.07063527 0.01872968]
 [ 0.39009179 0.27203897 -0.4711558 -0.14891121]
 [ 0.29615352 0.03811049 0.07930682 0.16183425]], shape=(4, 4), dtype=float32)
PS C:\Users\ayu\OneDrive\AI\BAG 4>

```

Program memvisualisasikan feature maps dari convolutional layer VGG19. Feature maps memperlihatkan fitur yang berhasil dideteksi oleh setiap filter pada gambar.

Example 4.19



2020-09-30 20:14:58.330811: I tensorflow/core/platform/cpu_feature_guard.cc:118] This TensorFlow binary is optimized to use available CPU instructions in performance-critical operations. To enable the following instructions: SSE3 SSE4.1 SSE4.2 AVX AVX2 FMA, in other operations, rebuild TensorFlow with the appropriate compiler flags.

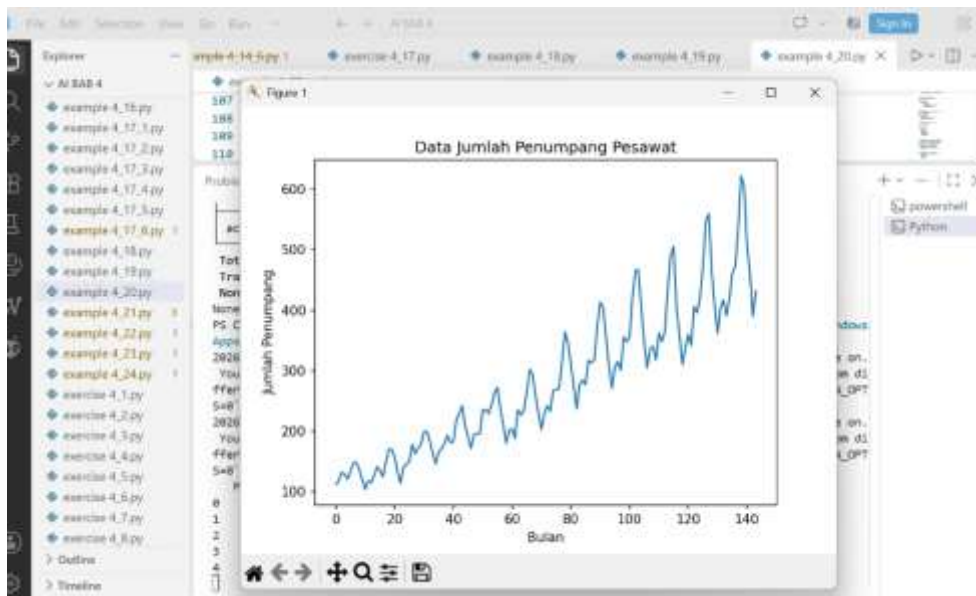
C:\Users\ayu zenska\AppData\Local\Packages\PythonSoftwareFoundation.Python.3.9_ghz5n2kfra8p0\LocalCache\local-packages\Python39\site-packages\keras\src\layers\rnn\rnn.py:199: UserWarning: Do not pass an 'input_shape' / 'input_dim' argument to a layer. When using Sequential models, prefer using an 'Input(shape)' object as the first layer in the model instead.
super().__init__(**kwargs)
Model: "sequential"

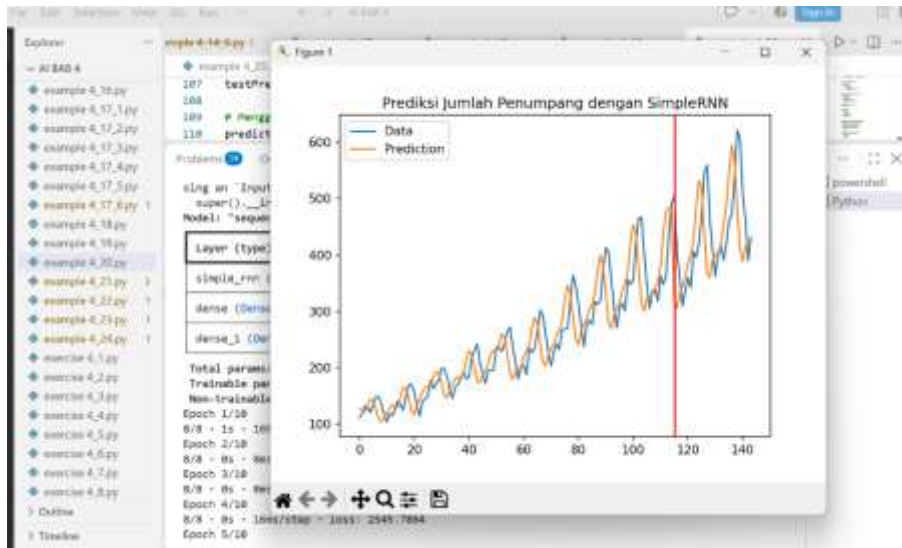
Layer (type)	Output Shape	Param #
simpla_rnn (SimpleRNN)	(None, 50)	2,000
dense (Dense)	(None, 46)	2,346
activation (Activation)	(None, 46)	0

Total params: 4,346 (19.32 KB)
Trainable params: 4,346 (19.32 KB)
Non-trainable params: 0 (0.00 B)
None
PS C:\Users\ayu zenska\OneDrive\AI BAB 4>

Program melanjutkan visualisasi feature maps pada layer yang berbeda. Layer yang lebih dalam menghasilkan fitur yang cenderung lebih kompleks.

Example 4.20





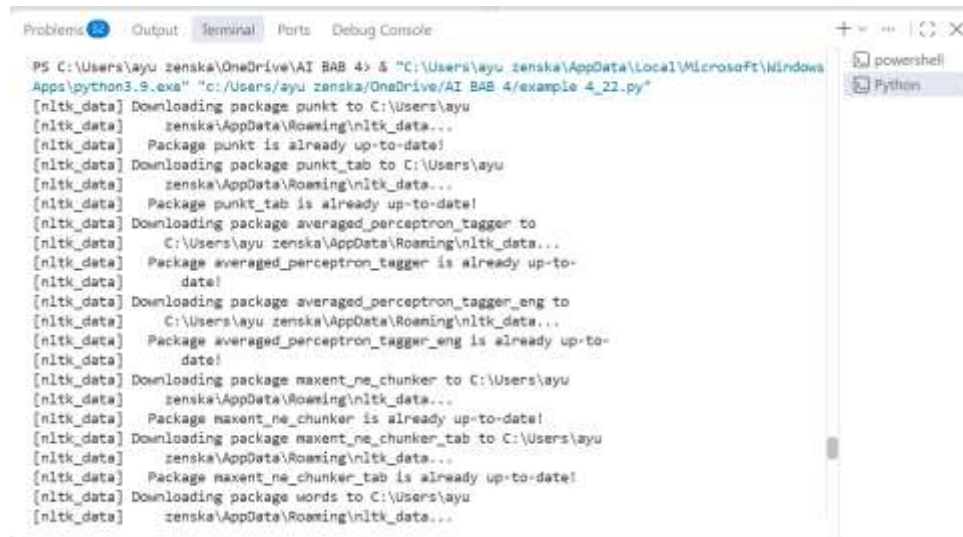
SimpleRNN digunakan untuk memprediksi jumlah penumpang pesawat berdasarkan data sebelumnya. Hasil menunjukkan bahwa RNN dapat mempelajari pola pada data yang memiliki urutan waktu.

Example 4.21



Program memperkenalkan pemrosesan teks menggunakan library NLP. Data teks diolah agar dapat dianalisis berdasarkan struktur dan informasi yang terkandung di dalamnya.

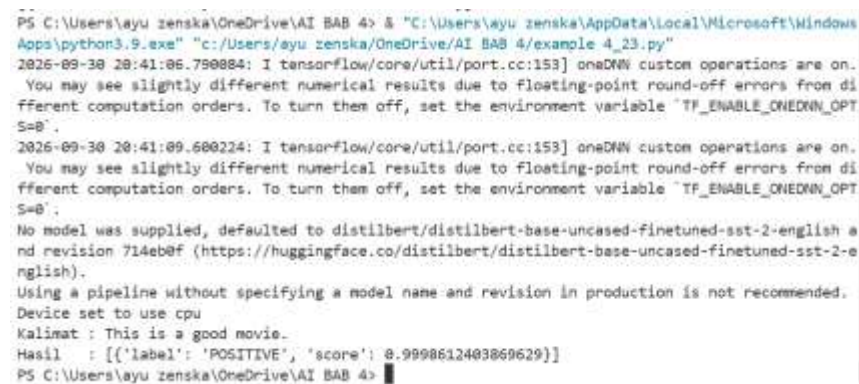
Example 4.22



```
PS C:\Users\ayu zenska\OneDrive\AI BAB 4> & "C:\Users\ayu zenska\AppData\Local\Microsoft\Windows  
Apps\python3.9.exe" "c:/Users/ayu zenska/OneDrive/AI BAB 4/example_4_22.py"  
[nltk_data] Downloading package punkt to C:\Users\ayu  
[nltk_data] zenska\AppData\Roaming\nltk_data...  
[nltk_data] Package punkt is already up-to-date!  
[nltk_data] Downloading package punkt_tab to C:\Users\ayu  
[nltk_data] zenska\AppData\Roaming\nltk_data...  
[nltk_data] Package punkt_tab is already up-to-date!  
[nltk_data] Downloading package averaged_perceptron_tagger to  
[nltk_data] C:\Users\ayu zenska\AppData\Roaming\nltk_data...  
[nltk_data] Package averaged_perceptron_tagger is already up-to-  
[nltk_data] date!  
[nltk_data] Downloading package averaged_perceptron_tagger_eng to  
[nltk_data] C:\Users\ayu zenska\AppData\Roaming\nltk_data...  
[nltk_data] Package averaged_perceptron_tagger_eng is already up-to-  
[nltk_data] date!  
[nltk_data] Downloading package maxent_ne_chunker to C:\Users\ayu  
[nltk_data] zenska\AppData\Roaming\nltk_data...  
[nltk_data] Package maxent_ne_chunker is already up-to-date!  
[nltk_data] Downloading package maxent_ne_chunker_tab to C:\Users\ayu  
[nltk_data] zenska\AppData\Roaming\nltk_data...  
[nltk_data] Package maxent_ne_chunker_tab is already up-to-date!  
[nltk_data] Downloading package words to C:\Users\ayu  
[nltk_data] zenska\AppData\Roaming\nltk_data...
```

NLTK digunakan untuk tokenisasi, POS tagging, dan Named Entity Recognition. Program dapat memecah teks, mengenali jenis kata, serta menemukan entitas tertentu.

Example 4.23



```
PS C:\Users\ayu zenska\OneDrive\AI BAB 4> & "C:\Users\ayu zenska\AppData\Local\Microsoft\Windows  
Apps\python3.9.exe" "c:/Users/ayu zenska/OneDrive/AI BAB 4/example_4_23.py"  
2026-09-30 20:41:06.790884: I tensorflow/core/util/port.cc:153] oneDNN custom operations are on.  
You may see slightly different numerical results due to floating-point round-off errors from di  
fferent computation orders. To turn them off, set the environment variable 'TF_ENABLE_ONEDNN_OPT  
S=0'.  
2026-09-30 20:41:09.680224: I tensorflow/core/util/port.cc:153] oneDNN custom operations are on.  
You may see slightly different numerical results due to floating-point round-off errors from di  
fferent computation orders. To turn them off, set the environment variable 'TF_ENABLE_ONEDNN_OPT  
S=0'.  
No model was supplied, defaulted to distilbert/distilbert-base-uncased-finetuned-sst-2-english a  
nd revision 714eb0f (https://huggingface.co/distilbert/distilbert-base-uncased-finetuned-sst-2-english).  
Using a pipeline without specifying a model name and revision in production is not recommended.  
Device set to use cpu  
Kalimat : This is a good movie.  
Hasil : [{'label': 'POSITIVE', 'score': 0.9998612403869629}]  
PS C:\Users\ayu zenska\OneDrive\AI BAB 4>
```

Transformers digunakan untuk melakukan sentiment analysis. Model dapat mengklasifikasikan teks menjadi sentimen positif atau negatif beserta tingkat keyakinannya.

Example 4.24

```
PS C:\Users\ayu zenska\OneDrive\AI BAB 4> & "C:\Users\ayu zenska\AppData\Local\Microsoft\Windows
Apps\python3.9.exe" "c:/Users/ayu zenska/OneDrive/AI BAB 4/example_4_24.py"
2026-09-30 20:47:02.534935: I tensorflow/core/util/port.cc:153] oneDNN custom operations are on.
You may see slightly different numerical results due to floating-point round-off errors from di
fferent computation orders. To turn them off, set the environment variable 'TF_ENABLE_ONEDNN_OPT
S=0'.
2026-09-30 20:47:05.651689: I tensorflow/core/util/port.cc:153] oneDNN custom operations are on.
You may see slightly different numerical results due to floating-point round-off errors from di
fferent computation orders. To turn them off, set the environment variable 'TF_ENABLE_ONEDNN_OPT
S=0'.
No model was supplied, defaulted to distilbert/distilbert-base-cased-distilled-squad and revisio
n 564e9b5 (https://huggingface.co/distilbert/distilbert-base-cased-distilled-squad).
Using a pipeline without specifying a model name and revision in production is not recommended.
WARNING:tensorflow:From C:\Users\ayu zenska\AppData\Local\Packages\PythonSoftwareFoundation.Pyth
on.3.9_qbz5n2kfra8p0\LocalCache\local-packages\Python39\site-packages\tf_keras\src\losses.py:297
6: The name tf.losses.sparse_softmax_cross_entropy is deprecated. Please use tf.compat.v1.losses
.sparse_softmax_cross_entropy instead.

Device set to use cpu:
C:\Users\ayu zenska\AppData\Local\Packages\PythonSoftwareFoundation.Python.3.9_qbz5n2kfra8p0\Loc
alCache\local-packages\Python39\site-packages\transformers\pipelines\question_answering.py:395:
FutureWarning: Passing a list of SQuAD examples to the pipeline is deprecated and will be remove
d in v5. Inputs should be passed using the 'question' and 'context' keyword arguments instead.
warnings.warn(
```

Pipeline question-answering digunakan untuk menemukan jawaban dari sebuah pertanyaan berdasarkan informasi yang terdapat dalam teks. Model mengambil bagian teks yang dianggap paling sesuai dengan pertanyaan.

➤ Exercise : Exercise 4.1

```
PS C:\Users\ayu zenska\OneDrive\AI BAB 4> & "C:\Users\ayu zenska\AppData\Local\Microsoft\Windows
Apps\python3.9.exe" "c:/Users/ayu zenska/OneDrive/AI BAB 4/exercise_4_1.py"
[0.9968177]
PS C:\Users\ayu zenska\OneDrive\AI BAB 4> █
```

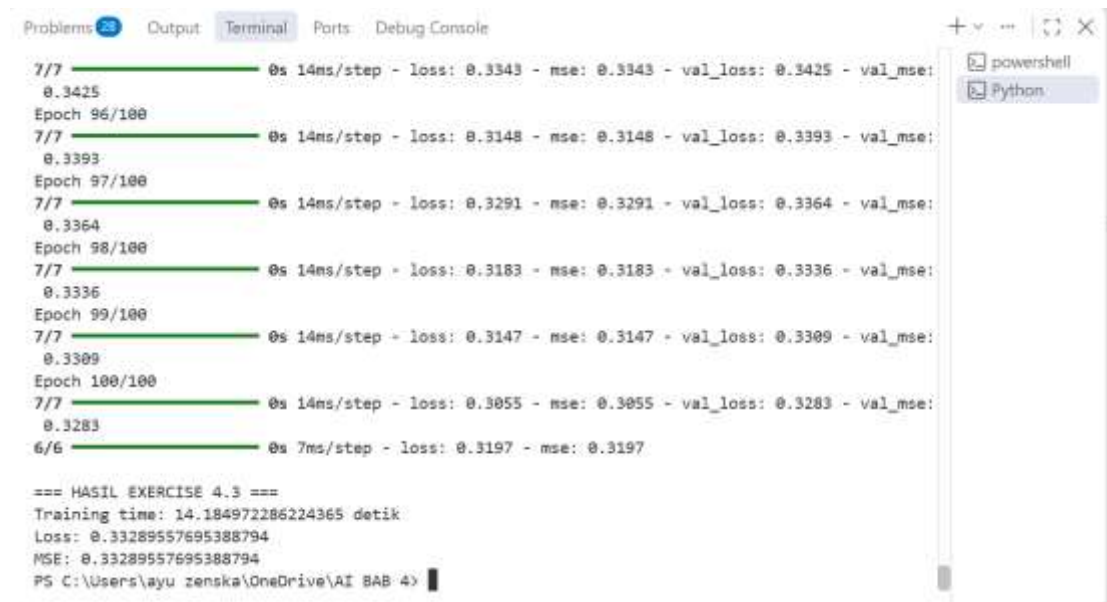
Model dari contoh sebelumnya dimodifikasi untuk menguji perubahan pada input atau proses pembelajaran. Hasil menunjukkan bahwa perubahan parameter dapat memengaruhi proses pembelajaran model.

Exercise 4.2

```
PS C:\Users\ayu zenska\OneDrive\AI BAB 4> & "C:\Users\ayu zenska\AppData\Local\Microsoft\Windows
Apps\python3.9.exe" "c:/Users/ayu zenska/OneDrive/AI BAB 4/exercise_4_2.py"
[1 0]
[(3, 5), (5, 2), (2, 1)]
PS C:\Users\ayu zenska\OneDrive\AI BAB 4> █
```

Jaringan dengan jumlah input atau neuron yang berbeda diuji untuk melihat pengaruh perubahan struktur terhadap hasil model.

Exercise 4.3



```
7/7 ██████████ 0s 14ms/step - loss: 0.3343 - mse: 0.3343 - val_loss: 0.3425 - val_mse: 0.3425
Epoch 96/100
7/7 ██████████ 0s 14ms/step - loss: 0.3148 - mse: 0.3148 - val_loss: 0.3393 - val_mse: 0.3393
Epoch 97/100
7/7 ██████████ 0s 14ms/step - loss: 0.3291 - mse: 0.3291 - val_loss: 0.3364 - val_mse: 0.3364
Epoch 98/100
7/7 ██████████ 0s 14ms/step - loss: 0.3183 - mse: 0.3183 - val_loss: 0.3336 - val_mse: 0.3336
Epoch 99/100
7/7 ██████████ 0s 14ms/step - loss: 0.3147 - mse: 0.3147 - val_loss: 0.3309 - val_mse: 0.3309
Epoch 100/100
7/7 ██████████ 0s 14ms/step - loss: 0.3055 - mse: 0.3055 - val_loss: 0.3283 - val_mse: 0.3283
6/6 ██████████ 0s 7ms/step - loss: 0.3197 - mse: 0.3197

=== HASIL EXERCISE 4.3 ===
Training time: 14.184972286224365 detik
Loss: 0.33289557695388794
MSE: 0.33289557695388794
PS C:\Users\ayu zenska\OneDrive\AI BAB 4>
```

Model MLP dimodifikasi dengan perubahan arsitektur. Hasil menunjukkan bahwa struktur layer dapat memengaruhi kemampuan jaringan dalam mempelajari data.

Exercise 4.4



Layer (type)	Output Shape	Param #
conv2d (Conv2D)	(None, 30, 30, 8)	80
average_pooling2d (AveragePooling2D)	(None, 15, 15, 8)	0
conv2d_1 (Conv2D)	(None, 13, 13, 20)	1,460
average_pooling2d_1 (AveragePooling2D)	(None, 6, 6, 20)	0
flatten (Flatten)	(None, 720)	0
dense (Dense)	(None, 120)	86,520
dense_1 (Dense)	(None, 84)	10,164
dense_2 (Dense)	(None, 10)	850

Total params: 99,074 (387.01 KB)
Trainable params: 99,074 (387.01 KB)
Non-trainable params: 0 (0.00 KB)
PS C:\Users\ayu zenska\OneDrive\AI BAB 4>

Jumlah filter pada CNN diubah. Perubahan jumlah filter menghasilkan jumlah parameter dan kemampuan ekstraksi fitur yang berbeda.

Exercise 4.5



Layer (type)	Output Shape	Param #
conv2d (Conv2D)	(None, 54, 54, 96)	34,944
max_pooling2d (MaxPooling2D)	(None, 27, 27, 96)	0
conv2d_1 (Conv2D)	(None, 17, 17, 256)	2,573,952
max_pooling2d_1 (MaxPooling2D)	(None, 8, 8, 256)	0
conv2d_2 (Conv2D)	(None, 6, 6, 384)	885,120
conv2d_3 (Conv2D)	(None, 4, 4, 384)	1,327,488
conv2d_4 (Conv2D)	(None, 2, 2, 256)	884,992
max_pooling2d_2 (MaxPooling2D)	(None, 1, 1, 256)	0
Flatten (Flatten)	(None, 256)	0
dense (Dense)	(None, 2048)	526,336
dropout (Dropout)	(None, 2048)	0

Arsitektur CNN dimodifikasi untuk melihat pengaruh perubahan layer terhadap hasil klasifikasi gambar.

Exercise 4.6

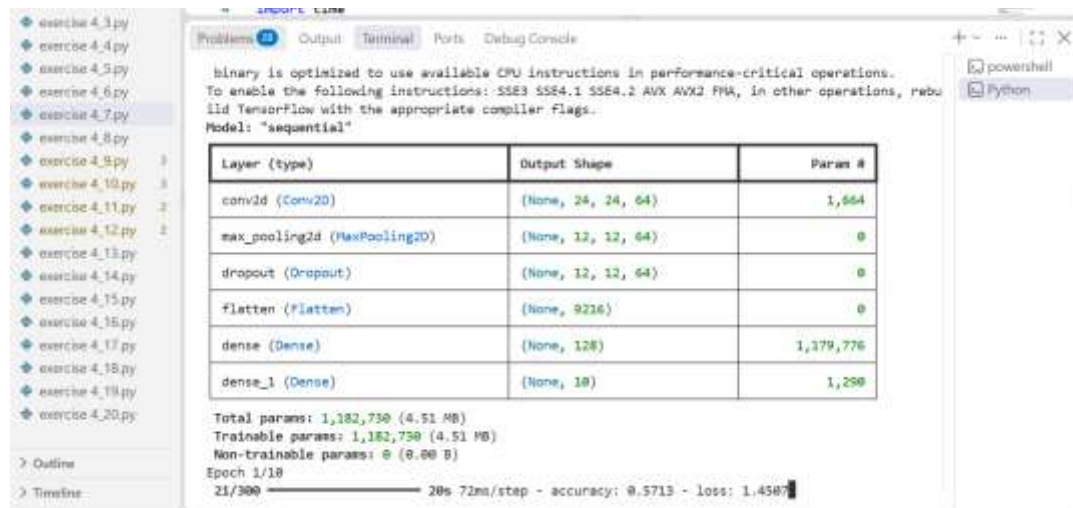


Layer (type)	Output Shape	Param #
conv2d (Conv2D)	(None, 24, 24, 32)	832
max_pooling2d (MaxPooling2D)	(None, 12, 12, 32)	0
dropout (Dropout)	(None, 12, 12, 32)	0
conv2d_1 (Conv2D)	(None, 10, 10, 64)	18,496
max_pooling2d_1 (MaxPooling2D)	(None, 5, 5, 64)	0
dropout_1 (Dropout)	(None, 5, 5, 64)	0
Flatten (Flatten)	(None, 1600)	0
dense (Dense)	(None, 128)	204,928
dense_1 (Dense)	(None, 2)	258

Total params: 224,514 (877.01 KB)
Trainable params: 224,514 (877.01 KB)
Non-trainable params: 0 (0.00 KB)
None
P5 C:\Users\ayu zenska\OneDrive\AI BAB 4>

Model CNN diuji dengan perubahan parameter tertentu. Hasil menunjukkan bahwa konfigurasi model berpengaruh terhadap proses pembelajaran.

Exercise 4.7



binary is optimized to use available CPU instructions in performance-critical operations.
To enable the following instructions: SSE3 SSE4.1 SSE4.2 AVX AVX2 FMA, in other operations, rebuild TensorFlow with the appropriate compiler flags.

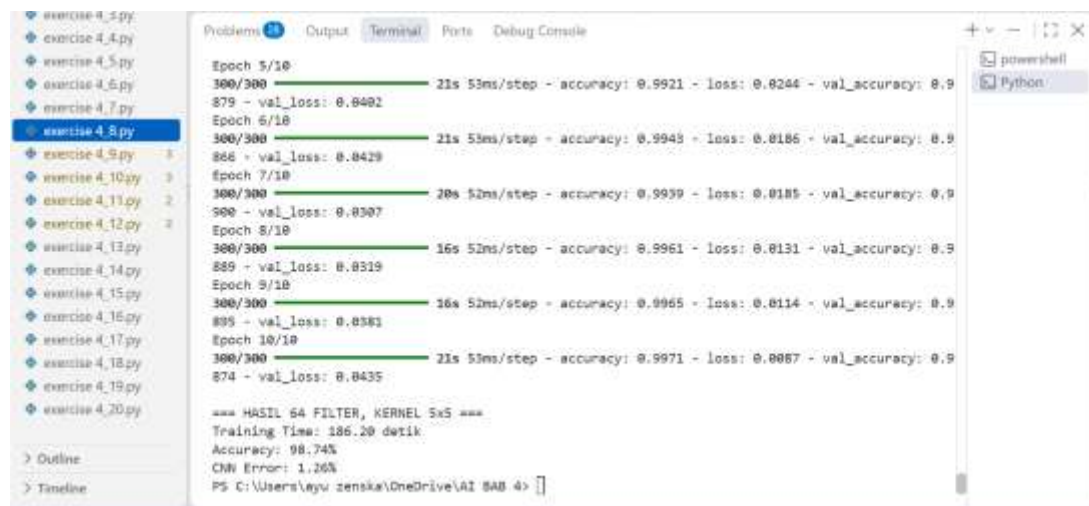
Model: "sequential"

Layer (type)	Output Shape	Param #
conv2d (Conv2D)	(None, 24, 24, 64)	1,664
max_pooling2d (MaxPooling2D)	(None, 12, 12, 64)	0
dropout (Dropout)	(None, 12, 12, 64)	0
flatten (Flatten)	(None, 9216)	0
dense (Dense)	(None, 128)	1,179,776
dense_1 (Dense)	(None, 10)	1,290

Total params: 1,182,730 (4.51 MB)
Trainable params: 1,182,730 (4.51 MB)
Non-trainable params: 0 (0.00 B)
Epoch 1/10
21/300 ————— 20s 72ms/step - accuracy: 0.5713 - loss: 1.4587

Penambahan convolutional dan pooling layer membuat model dapat melakukan ekstraksi fitur secara lebih bertahap dari gambar.

Exercise 4.8

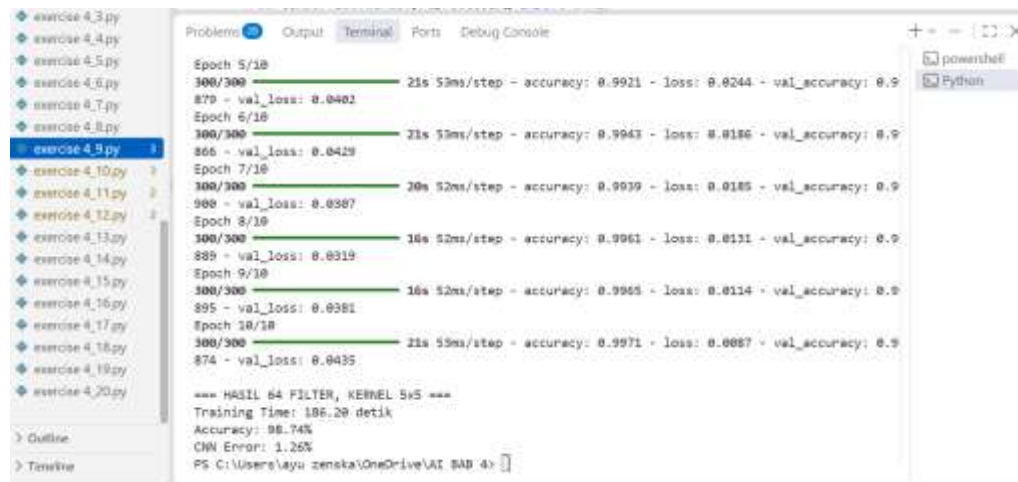


Epoch 5/10
300/300 ————— 21s 53ms/step - accuracy: 0.9921 - loss: 0.0244 - val_accuracy: 0.9
879 - val_loss: 0.0402
Epoch 6/10
300/300 ————— 21s 53ms/step - accuracy: 0.9945 - loss: 0.0186 - val_accuracy: 0.9
866 - val_loss: 0.0429
Epoch 7/10
300/300 ————— 20s 52ms/step - accuracy: 0.9959 - loss: 0.0185 - val_accuracy: 0.9
900 - val_loss: 0.0307
Epoch 8/10
300/300 ————— 16s 52ms/step - accuracy: 0.9961 - loss: 0.0131 - val_accuracy: 0.9
889 - val_loss: 0.0319
Epoch 9/10
300/300 ————— 16s 52ms/step - accuracy: 0.9965 - loss: 0.0114 - val_accuracy: 0.9
895 - val_loss: 0.0583
Epoch 10/10
300/300 ————— 21s 53ms/step - accuracy: 0.9971 - loss: 0.0087 - val_accuracy: 0.9
874 - val_loss: 0.0435

=== HASIL 64 FILTER, KERNEL 5x5 ===
Training Time: 186.20 detik
Accuracy: 98.74%
CNN Error: 1.26%
PS C:\Users\ayu_zenska\OneDrive\AI BAB 4>

Model diuji dengan konfigurasi CNN yang berbeda. Perubahan arsitektur dapat menghasilkan karakteristik prediksi yang berbeda.

Exercise 4.9



```
Exercise 4.3.py
Exercise 4.4.py
Exercise 4.5.py
Exercise 4.6.py
Exercise 4.7.py
Exercise 4.8.py
Exercise 4.9.py
Exercise 4.10.py
Exercise 4.11.py
Exercise 4.12.py
Exercise 4.13.py
Exercise 4.14.py
Exercise 4.15.py
Exercise 4.16.py
Exercise 4.17.py
Exercise 4.18.py
Exercise 4.19.py
Exercise 4.20.py

> Outline
> Timeline

Epoch 5/10
300/300 ————— 21s 53ms/step - accuracy: 0.9921 - loss: 0.0244 - val_accuracy: 0.9
879 - val_loss: 0.0402
Epoch 6/10
300/300 ————— 21s 53ms/step - accuracy: 0.9943 - loss: 0.0186 - val_accuracy: 0.9
866 - val_loss: 0.0429
Epoch 7/10
300/300 ————— 20s 52ms/step - accuracy: 0.9939 - loss: 0.0185 - val_accuracy: 0.9
900 - val_loss: 0.0387
Epoch 8/10
300/300 ————— 16s 52ms/step - accuracy: 0.9961 - loss: 0.0131 - val_accuracy: 0.9
889 - val_loss: 0.0319
Epoch 9/10
300/300 ————— 16s 52ms/step - accuracy: 0.9965 - loss: 0.0114 - val_accuracy: 0.9
895 - val_loss: 0.0381
Epoch 10/10
300/300 ————— 21s 53ms/step - accuracy: 0.9971 - loss: 0.0087 - val_accuracy: 0.9
874 - val_loss: 0.0435

=== HASIL 64 FILTER, KERNEL 5x5 ===
Training Time: 186.29 detik
Accuracy: 98.74%
CNN Error: 1.26%
PS C:\Users\ayu zenska\OneDrive\AI BAB 4>
```

Model klasifikasi gambar dimodifikasi dan hasilnya dibandingkan dengan model sebelumnya. Hal ini menunjukkan pengaruh perubahan arsitektur terhadap hasil klasifikasi.

Exercise 4.10



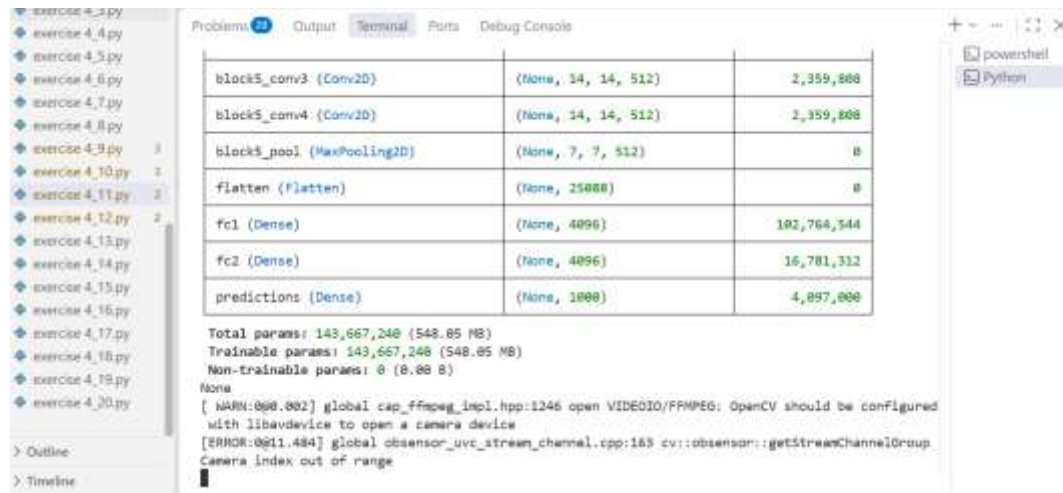
```
Exercise 4.3.py
Exercise 4.4.py
Exercise 4.5.py
Exercise 4.6.py
Exercise 4.7.py
Exercise 4.8.py
Exercise 4.9.py
Exercise 4.10.py
Exercise 4.11.py
Exercise 4.12.py
Exercise 4.13.py
Exercise 4.14.py
Exercise 4.15.py
Exercise 4.16.py
Exercise 4.17.py
Exercise 4.18.py
Exercise 4.19.py
Exercise 4.20.py

> Outline
> Timeline

=== HASIL 64 FILTER, KERNEL 5x5 ===
Training Time: 186.29 detik
Accuracy: 98.74%
CNN Error: 1.26%
PS C:\Users\ayu zenska\OneDrive\AI BAB 4> & "C:\Users\ayu zenska\AppData\Local\Microsoft\Windows
Apps\python3.9.exe" "C:\Users\ayu zenska\OneDrive\AI BAB 4\exercise 4.10.py"
2026-09-30 20:56:07.534583: I tensorflow/core/util/port.cc:153] oneDNN custom operations are on.
You may see slightly different numerical results due to floating-point round-off errors from di
fferent computation orders. To turn them off, set the environment variable 'TF_ENABLE_ONEDNN_OPT
5=0'.
2026-09-30 20:56:10.792095: I tensorflow/core/util/port.cc:153] oneDNN custom operations are on.
You may see slightly different numerical results due to floating-point round-off errors from di
fferent computation orders. To turn them off, set the environment variable 'TF_ENABLE_ONEDNN_OPT
5=0'.
2026-09-30 20:56:11.705852: I tensorflow/core/platform/cpu_feature_guard.cc:210] This TensorFlow
binary is optimized to use available CPU instructions in performance-critical operations.
To enable the following instructions: SSE4.1 SSE4.2 AVX AVX2 FMA, in other operations, rebu
ild TensorFlow with the appropriate compiler flags.
1/1 ————— 1s 98ms/step
[[('n01871265', 'Eusker', 0.5571216), ('n02584815', 'Indian_elephant', 0.23468867), ('n02584458',
'African_elephant', 0.28773214), ('n02092329', 'Maimaraheh', 0.622414e-05), ('n01408429', 'wat
er_buffalo', 5.2763004e-05)]]
PS C:\Users\ayu zenska\OneDrive\AI BAB 4>
```

VGG19 digunakan untuk mengklasifikasikan gambar yang sama dengan VGG16. Hasil prediksi dapat berbeda karena kedua model memiliki struktur dan bobot yang berbeda.

Exercise 4.11



Layer	Shape	Params
block5_conv3 (Conv2D)	(None, 14, 14, 512)	2,359,808
block5_conv4 (Conv2D)	(None, 14, 14, 512)	2,359,808
block5_pool (MaxPooling2D)	(None, 7, 7, 512)	0
flatten (Flatten)	(None, 25088)	0
fc1 (Dense)	(None, 4096)	102,764,344
fc2 (Dense)	(None, 4096)	16,781,312
predictions (Dense)	(None, 1000)	4,097,000

Total params: 143,667,240 (548.05 MB)
Trainable params: 143,667,240 (548.05 MB)
Non-trainable params: 0 (0.00 B)

None
[WARN:090.892] global cap_ffmpeg_impl.hpp:1246 open VIDEOIO/FFMPEG: OpenCV should be configured with libavdevice to open a camera device
[ERROR:0911.484] global obsensor_uvc_stream_channel.cpp:163 cv::obsensor::getStreamChannelGroup: Camera index out of range

Model CNN digunakan dengan input dari webcam. Hasil klasifikasi menunjukkan bahwa model dapat memproses gambar secara langsung.

Exercise 4.12



Layer	Shape	Params	Layer
conv5_block16_1_conv (Conv2D)	(None, 7, 7, 32)	36,864	conv5_block16_1_relu...
conv5_block16_concat (Concatenate)	(None, 7, 7, 1024)	0	conv5_block16_concat...
bn (BatchNormalization)	(None, 7, 7, 1024)	4,096	conv5_block16_concat...
relu (Activation)	(None, 7, 7, 1024)	0	bn[0][0]
avg_pool (GlobalAveragePooling2D)	(None, 1024)	0	relu[0][0]
predictions (Dense)	(None, 1000)	1,025,000	avg_pool[0][0]

Total params: 8,062,504 (30.75 MB)
Trainable params: 7,078,016 (30.44 MB)
Non-trainable params: 83,648 (326.75 KB)

None
[WARN:090.000] global cap_ffmpeg_impl.hpp:1246 open VIDEOIO/FFMPEG: OpenCV should be configured with libavdevice to open a camera device
[ERROR:090.620] global obsensor_uvc_stream_channel.cpp:163 cv::obsensor::getStreamChannelGroup: Camera index out of range
PS C:\Users\ayu\OneDrive\AI BAB 4>

Model CNN lain seperti DenseNet digunakan sebagai pembanding. Setiap arsitektur dapat menghasilkan prediksi dan confidence yang berbeda.

Exercise 4.13

The screenshot shows a Jupyter Notebook interface. On the left, a list of exercises is visible, with 'exercise 4.13.py' selected. The main area displays a table of model layers and their parameters:

Layer	Shape	Params
block5_conv2 (Conv2D)	(None, 14, 14, 512)	2,359,808
block5_conv3 (Conv2D)	(None, 14, 14, 512)	2,359,808
block5_pool (MaxPooling2D)	(None, 7, 7, 512)	0
flatten (Flatten)	(None, 25088)	0
fc1 (Dense)	(None, 4096)	102,764,544
fc2 (Dense)	(None, 4096)	16,781,312
predictions (Dense)	(None, 1000)	4,097,000

Below the table, the following text is displayed:

```
Total params: 138,357,544 (527.79 MB)
Trainable params: 138,357,544 (527.79 MB)
Non-trainable params: 0 (0.00 B)
[ WARN:000.005] global cap_ffmpeg_impl.hpp:1246 open VIDEOIO/FFMPEG: OpenCV should be configured
with libavdevice to open a camera device
[ERROR:008.504] global obsensor_uvc_stream_channel.cpp:163 cv::obsensor::getStreamChannelGroup C
amera index out of range
Tidak dapat membaca kamera.
PS C:\Users\ayu zenska\OneDrive\AI BAB 4>
```

On the right, a terminal window shows the command to run the script:

```
PS C:\Users\ayu zenska\OneDrive\AI BAB 4> & "C:\Users\ayu zenska\AppData\local\Microsoft\Windows
Apps\python3.9.exe" "c:/Users/ayu zenska/OneDrive/AI BAB 4/exercise 4.14.py"
```

Autoencoder dimodifikasi untuk melihat pengaruh perubahan parameter terhadap hasil rekonstruksi gambar.

Exercise 4.14

The screenshot shows a Jupyter Notebook interface. On the left, a list of exercises is visible, with 'exercise 4.14.py' selected. The main area displays a list of model choices:

```
Pilih model:
1. VGG16
2. ResNet50
3. MobileNetV2
4. DenseNet201
5. InceptionV3
Masukkan pilihan (1-5): 2
```

Below the list, the following text is displayed:

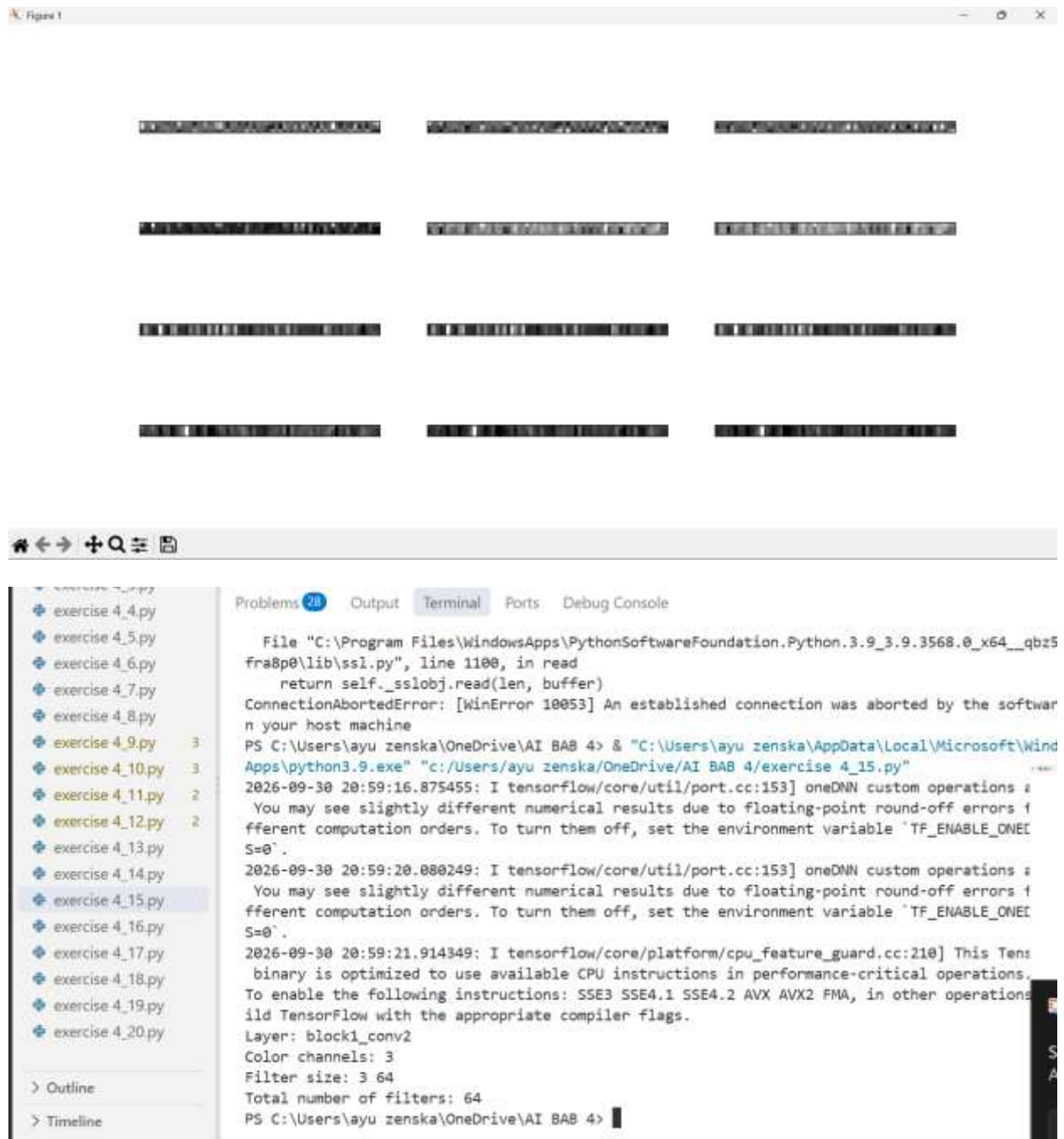
```
Model yang dipilih: ResNet50
2026-09-30 20:58:37.783269: I tensorflow/core/platform/cpu_feature_guard.cc:210] This TensorFlow
binary is optimized to use available CPU instructions in performance-critical operations.
To enable the following instructions: SSE3 SSE4.1 SSE4.2 AVX AVX2 FMA, in other operations, rebu
ild TensorFlow with the appropriate compiler flags.
Downloading data from https://github.com/qubvel/classification_models/releases/download/0.0.1/re
```

On the right, a terminal window shows the command to run the script:

```
PS C:\Users\ayu zenska\OneDrive\AI BAB 4> & "C:\Users\ayu zenska\AppData\local\Microsoft\Windows
Apps\python3.9.exe" "c:/Users/ayu zenska/OneDrive/AI BAB 4/exercise 4.14.py"
```

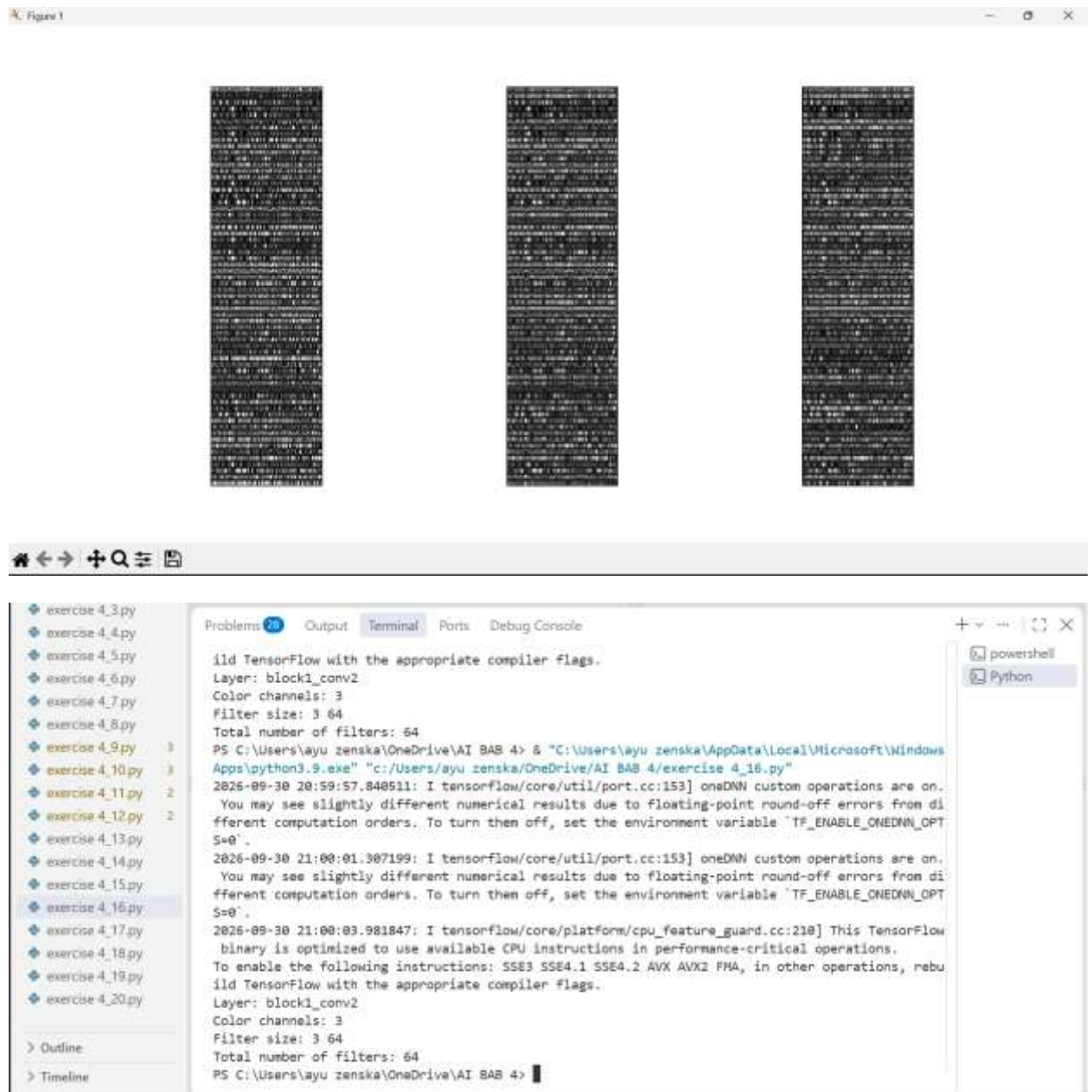
Hasil rekonstruksi dan nilai loss diamati untuk mengetahui kemampuan model dalam mempertahankan informasi gambar setelah proses kompresi.

Exercise 4.15



Convolutional layer kedua VGG19 divisualisasikan. Feature maps mulai menunjukkan pola yang lebih beragam dibandingkan layer pertama.

Exercise 4.16



Seluruh filter pada convolutional layer divisualisasikan. Hasil menunjukkan adanya berbagai pola yang digunakan CNN untuk mengekstraksi fitur gambar.

Exercise 4.17



The figure displays a grid of 80 small images, arranged in 8 columns and 10 rows. Each column represents a different input image, and each row represents a different feature map. The feature maps show various patterns and textures extracted from the input images.

The interface below the grid shows a list of exercises on the left, with 'exercise 4.9.py' selected. The main area displays the output of the exercise, including a table of layer parameters and a summary of the model's performance.

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You may see slightly different numerical results due to floating-point round-off errors from different computation orders. To turn them off, set the environment variable 'TF_ENABLE_ONEDNN_OPTS=0'.

2026-09-30 21:00:55.478718: I tensorflow/core/platform/cpu_feature_guard.cc:210] This TensorFlow binary is optimized to use available CPU instructions in performance-critical operations.
To enable the following instructions: SSE3 SSE4.1 SSE4.2 AVX AVX2 FMA, in other operations, rebuild TensorFlow with the appropriate compiler flags.

Layer yang digunakan: block1_conv2
Model: "functional"

Layer (type)	Output Shape	Param #
input_layer (InputLayer)	(None, 224, 224, 3)	0
block1_conv1 (Conv2D)	(None, 224, 224, 64)	1,792
block1_conv2 (Conv2D)	(None, 224, 224, 64)	36,928

Total params: 38,720 (151.25 KB)
Trainable params: 38,720 (151.25 KB)
Non-trainable params: 0 (0.00 B)
1/1 ----- 0s 34ms/step
Feature maps: (1, 224, 224, 64)
PS C:\Users\ayu zenska\OneDrive\AI BAB 4>

Beberapa convolutional layer VGG19 dipilih untuk melihat perubahan feature maps. Semakin dalam layer, fitur yang dipelajari cenderung semakin kompleks.

Exercise 4.18



```
exercise 4_3.py
exercise 4_4.py
exercise 4_5.py
exercise 4_6.py
exercise 4_7.py
exercise 4_8.py
exercise 4_9.py 3
exercise 4_10.py 3
exercise 4_11.py 2
exercise 4_12.py 2
exercise 4_13.py
exercise 4_14.py
exercise 4_15.py
exercise 4_16.py
exercise 4_17.py
exercise 4_18.py
exercise 4_19.py
exercise 4_20.py

> Outline
> Timeline

Problems Output Terminal Ports Debug Console

((
(ORGANIZATION PENS/NNP)
))
is/VBZ
a/DT
university/NN
in/IN
(GPE Surabaya/NNP)
(GPE Indonesia/NNP)
Students/NN
learn/NN
technology/NN
multimedia/NN
engineering/NN
and/CC
artificial/NN
intelligence/NN
./.)
PS C:\Users\ayu zenska\OneDrive\AI BAB 4>
```

Teks yang berbeda dianalisis menggunakan NLTK. Tokenisasi, POS tagging, dan NER menunjukkan bahwa teks dapat diolah menjadi informasi yang lebih terstruktur.

Exercise 4.19



```
exercise 4_3.py
exercise 4_4.py
exercise 4_5.py
exercise 4_6.py
exercise 4_7.py
exercise 4_8.py
exercise 4_9.py 3
exercise 4_10.py 3
exercise 4_11.py 2
exercise 4_12.py 2
exercise 4_13.py
exercise 4_14.py
exercise 4_15.py
exercise 4_16.py
exercise 4_17.py
exercise 4_18.py
exercise 4_19.py
exercise 4_20.py

> Outline
> Timeline

Problems Output Terminal Ports Debug Console

on.3.9_qbz5n2kfra8p0\LocalCache\local-packages\Python39\site-packages\tf_keras\src\losses.py:297:
6: The name tf.losses.sparse_softmax_cross_entropy is deprecated. Please use tf.compat.v1.losses
.sparse_softmax_cross_entropy instead.

Device set to use cpu
Kalimat : I really love this movie.
Hasil : [{"label": 'POSITIVE', 'score': 0.9998745918273926}]

Kalimat : The food was delicious and amazing.
Hasil : [{"label": 'POSITIVE', 'score': 0.9998860359191895}]

Kalimat : This application is very useful.
Hasil : [{"label": 'POSITIVE', 'score': 0.9994286894798279}]

Kalimat : I hate this movie.
Hasil : [{"label": 'NEGATIVE', 'score': 0.9996869564056396}]

Kalimat : The food was terrible.
Hasil : [{"label": 'NEGATIVE', 'score': 0.999190869471741}]

Kalimat : This application is very difficult to use.
Hasil : [{"label": 'NEGATIVE', 'score': 0.9993045330047607}]

PS C:\Users\ayu zenska\OneDrive\AI BAB 4>
```

Beberapa kalimat positif dan negatif dianalisis menggunakan Transformers. Model dapat membedakan sentimen berdasarkan pola kata yang terdapat pada kalimat.

Exercise 4.20



```
base-cased-distilled-squad/resolve/main/tokenizer_config.json
Retrying in 8s [Retry 5/5].
'(MaxRetryError('HTTPSConnectionPool(host='huggingface.co', port=443): Max retries exceeded with url: /distilbert/distilbert-base-cased-distilled-squad/resolve/main/tokenizer_config.json (Caused by NameResolutionError("HTTPSConnection(host='huggingface.co', port=443): Failed to resolve 'huggingface.co' ([Errno 11001] getaddrinfo failed)"))', 'Request ID: 6edf96c6-f1f9-4b93-bcad-4ac3cbd6ffcf'))' thrown while requesting HEAD https://huggingface.co/distilbert/distilbert-base-cased-distilled-squad/resolve/main/tokenizer_config.json
Device set to use cpu
C:\Users\ayu zenska\AppData\Local\Packages\PythonSoftwareFoundation.Python.3.9_qbz5n2kfra8p0\LocalCache\local-packages\Python39\site-packages\transformers\pipelines\question_answering.py:395: FutureWarning: Passing a list of SQuAD examples to the pipeline is deprecated and will be removed in v5. Inputs should be passed using the 'question' and 'context' keyword arguments instead.
  warnings.warn(
Pertanyaan 1:
{'score': 0.8892405200004578, 'start': 75, 'end': 94, 'answer': 'Surabaya, Indonesia'}

Pertanyaan 2:
{'score': 0.4775364398956299, 'start': 128, 'end': 154, 'answer': 'technology and engineering'}

Pertanyaan 3:
{'score': 0.8665400743484497, 'start': 175, 'end': 223, 'answer': 'multimedia, \nrobotics, and information technology'}
PS C:\Users\ayu zenska\OneDrive\AI BAB 4> |
```

Teks dan pertanyaan diubah pada pipeline question-answering. Model dapat mengambil jawaban yang sesuai dari informasi yang tersedia pada teks.